

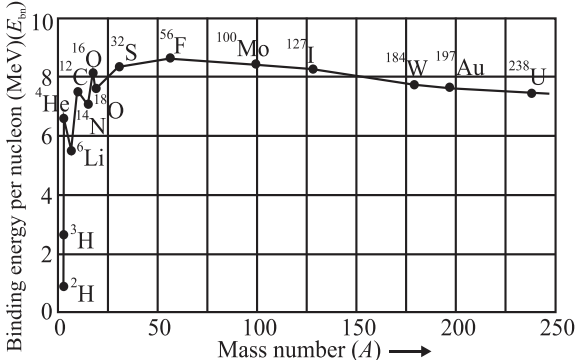


CHAPTER

13

Nuclei

Sr. No.	Concept	Formulas	Other information
1.	Atomic and mass number	Symbol of nucleus: ${}_Z X^A$ or ${}_Z^A X$ Atomic number (Z) = Number of protons Mass number (A) = $Z + N$ (N) = $A - Z$	N = number of neutrons X = symbol of element
2.	Size of nucleus	Volume, $V = \frac{4}{3}\pi R^3 \propto A \Rightarrow R \propto A^{1/3}$ $\Rightarrow R = R_0 A^{1/3} \Rightarrow R_0 = 1.2 \times 10^{-15}$	R = radius of nucleus A = mass number of the nucleus.
3.	Nuclear density	Density of nuclear matter = $\frac{\text{mass of nucleus}}{\text{volume of nucleus}}$ $\rho = \frac{mA}{\frac{4}{3}\pi R_0^3 A}$ or $\rho = \frac{3m}{4\pi R_0^3} = 2.38 \times 10^{17} \text{ kg m}^{-3}$	m = average mass of a nucleon = $1.67 \times 10^{-27} \text{ kg}$ A = mass number $R_0 = 1.2 \times 10^{-15} \text{ m}$
4.	Mass defect of nucleus	Δm = (Mass of protons + Mass of neutrons) – Mass of the nucleus $\Delta m = [Zm_p + (A - Z)m_n - m_N] \text{ a.m.u.}$ 1 a.m.u. (or 1u) = $1.66 \times 10^{-27} \text{ kg}$ Packing fraction = $\frac{\text{Actual mass of nucleus} - \text{Mass number}}{\text{Mass number}} = \frac{m - A}{A}$	Z = atomic number m_p = mass of a proton = 1.0073 a.m.u. = $1.673 \times 10^{-27} \text{ kg}$ m_n = mass of a neutron = 1.0087 a.m.u. = $1.675 \times 10^{-27} \text{ kg}$ m_N = mass of the nucleus
5.	Energy-mass relation	According to Einstein, $E = mc^2$	E = Energy equivalent of mass m c = speed of light
6.	Binding energy of nucleus	Binding energy $= \Delta mc^2 = [Zm_p + (A - Z)m_n - m_N] \cdot c^2$ If Δm is measured in a.m.u. $E_b = \Delta m \times 931 \text{ MeV}$ Where, $\Delta m = [Zm_p + (A - Z)m_n - m_N] \text{ a.m.u.}$ Binding energy per nucleon = $\frac{\Delta m \times 931 \text{ MeV}}{A}$	E_b = binding energy Δm = mass defect A = mass number 1 a.m.u. $\approx 931 \text{ MeV}$

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7.	Binding energy curve		E_{bn} = binding energy per nucleon
8.	Nuclear reaction	Nuclear fission; ${}_{92}^{235}\text{U} + {}_0^1\text{n} \longrightarrow {}_{56}^{141}\text{Ba} + {}_{36}^{92}\text{Kr} + 3{}_0^1\text{n} + Q.$ Nuclear fusion. ${}_1^2\text{H} + {}_1^2\text{H} \longrightarrow {}_2^4\text{He} + 24\text{MeV}.$	Q = Amount of heat released
9.	Q-Value	Energy of nuclear fission or fusion = (Mass of reactants – Mass of products) $\times$ 931 MeV	(Here all mass values are taken in a.m.u.)